Integrative and Comparative Biology

*Integrative and Comparative Biology*, volume 62, number 5, pp. 1320–1334
https://doi.org/10.1093/icb/icac047

*Society for Integrative and Comparative Biology*

SYMPOSIUM

# The Spring-Mass Model and Other Reductionist Models of Bipedal Locomotion on Inclines

Alessandro Maria Selvitella[*,†] and Kathleen Lois Foster[‡,1]

[*]Department of Mathematical Sciences, Purdue University Fort Wayne, 2101 E. Coliseum Blvd, Fort Wayne, 46805 IN, USA; [†]eScience Institute, University of Washington, 3910 15th Ave NE, Seattle, 98195 WA, USA; [‡]Department of Biology, Ball State University, 1600 Ashland Ave, 47306 IN, USA

From the symposium "Evolutionary conservation and diversity in a key vertebrate behavior: "walking" as a model system" presented at the annual meeting of the Society for Integrative and Comparative Biology virtual annual meeting, January 3–February 28, 2022.

[1]E-mail: klfoster@bsu.edu

**Synopsis** The spring-mass model is a model of locomotion aimed at giving the essential mathematical laws of the trajectory of the center of mass of an animal during bouncing gaits, such as hopping (one-dimensional) and running (two-dimensional). This reductionist mechanical system has been extensively investigated for locomotion over horizontal surfaces, whereas it has been largely neglected on other ecologically relevant surfaces, including inclines. For example, how the degree of inclination impacts the dynamics of the center of mass of the spring-mass model has not been investigated thoroughly. In this work, we derive a mathematical model which extends the spring-mass model to inclined surfaces. Among our results, we derive an approximate solution of the system, assuming a small angular sweep of the limb and a small spring compression during stance, and show that this approximation is very accurate, especially for small inclinations of the ground. Furthermore, we derive theoretical bounds on the difference between the Lagrangian and Lagrange equations of the true and approximate systems, and discuss locomotor stability questions of the approximate solutions. We test our models through a sensitivity analysis using parameters relevant to the locomotion of bipedal animals (quail, pheasant, guinea fowl, turkey, ostrich, and humans) and compare our approximate solution to the numerically derived solution of the exact system. We compare the two-dimensional spring-mass model on inclines with the one-dimensional spring-mass model to which it reduces under the limit of no horizontal velocity; we compare the two-dimensional spring-mass model on inclines with the inverted pendulum model on inclines towards which it converges in the case of high stiffness-to-mass ratio. We include comparisons with historically prevalent no-gravity approximations of these models, as well. The insights we have gleaned through all these comparisons and the ability of our approximation to replicate some of the kinematic changes observed in animals moving on different inclines (e.g., reduction in vertical oscillation of the center of mass and decreased stride length) underline the valuable and reasonable contributions that very simple, reductionist models, like the spring-mass model, can provide.

## Introduction

The world is full of extraordinary examples of terrestrial, aquatic, and aerial locomotion. These behaviors are critical for the survival and fitness of the animals that perform them, being essential for capturing prey, evading predators, and finding and competing for mates, and thus they play a key role in shaping the morphology and physiology of those animals (Garland and Losos 1994; Dickinson et al. 2000; Foster et al. 2015). Many locomotor behaviors are powered by paired appendages (fins, wings, or legs) pushing against the surfaces or medium around the body (Alexander 2003b; Gordon et al. 2017; Patek and Biewener 2018). Even within a strictly terrestrial environment, there is large morphological disparity both within and between bipedal, quadrupedal, and multipedal animals that execute a wide variety of locomotor patterns, including hoping, walking, and running (Alexander 2003b; Gordon et al. 2017; Patek and

Advance Access publication June 9, 2022
© The Author(s) 2022. Published by Oxford University Press on behalf of the Society for Integrative and Comparative Biology. All rights reserved.
For permissions, please e-mail: journals.permissions@oup.com

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

Biewener 2018). In the face of such morphological and biomechanical diversity, it is natural to wonder what elements are common across these legged systems. In other words, what fundamental principles drive legged locomotion?

There is a rich history of the use of models to answer this or similar questions (reviewed in Alexander 2003a): Mathematical models can be used to identify key variables, to find the conditions or features that optimize some behavior or performance characteristic, or to predict the outcome of manipulations of locomotor variables (e.g., Blickhan 1989; McMahon and Cheng 1990; Ghigliazza et al. 2003; Geyer et al. 2005; Holmes et al. 2006; Müller et al. 2016); physical models help to test and refine the principles proposed by mathematical models and to observe and study behaviors that are difficult to elicit in animals (e.g., McGeer 1990a, 1990b; Holmes et al. 2006; Iida et al. 2008; Raibert et al. 2008; Seok et al. 2013) and conceptual models strive to aid understanding, often through analogy, but without the rigor of mathematical deductions (e.g., Margaria 1976). In this manuscript, we will focus on the insights that simple mathematical models can provide towards our understanding of bipedal hopping, walking, and running.

Locomotor systems are highly complex, with multiple neural signals controlling the actions of multiple muscles applying forces across multiple joints in multiple limbs (Holmes et al. 2006; Gordon et al. 2017). This complexity requires complex integral representations of the dynamics. The identification of key components of such complex dynamics through mathematical modeling has been approached in two fundamentally opposite ways: first, through the successive removal of individual parameters or features from complex models in an effort to identify critical components without which models fail to resemble realistic behaviors; and second, through the building of models from fundamental principles and using only the most basic, indispensable components (Geyer 2005). The former approach has the advantage of having more components and, with that complexity, it is often able to more accurately reflect biological behaviors (Geyer 2005; Holmes et al. 2006). However, this complexity comes at a cost of interpretability, where the higher dimensionality makes it more difficult to identify likely culprits when models fail (Alexander 2003a; Geyer et al. 2006). In contrast, the latter approach can be criticized for being so simple that such models cannot accurately represent natural locomotor behaviors (Holmes et al. 2006); however, this simplicity is advantageous for identifying the parameters that lead to particular behaviors under study, allows for a more thorough examination of such parameters, and, once such parameters have been identified and characterized, they can be implemented as components of more complex, higher-dimensional models (Alexander 2003a; Geyer 2005). These "*reductionist models*" (RMs) are systems of differential equations that are so-named because they are reduced to include the dynamics of very few parameters of interest (e.g., center of mass; CoM). These RMs capture the most essential components of a locomotor behavior and have been used to provide numerous insights into fundamental principles that govern legged locomotion (Full and Koditschek 1999; Dickinson et al. 2000; Alexander 2003a; Geyer 2005).

The most common legged gaits that are described by RMs are hopping, walking, and running (Cavagna et al. 1977; Full and Koditschek 1999; Alexander 2003b; Patek and Biewener 2018; Gordon et al. 2017). Hopping on the spot (i.e., hopping with no horizontal movement) can be described through a single equation of motion, the one-dimensional *spring-mass model* (SMM) that determines the dynamics of the CoM with the vertical displacement from the contact surface (Farley et al. 1991; Seyfarth et al. 2001; Gordon et al. 2017). Walking and running are often described two-dimensionally through a system of differential equations for the $x$- and $y$-coordinates of the CoM in the sagittal plane of the body or through polar coordinates ($r$ for the radius and $\phi$ for the angle) (Alexander 1976; Mochon and Mcmahon 1980; Blickhan 1989; McMahon and Cheng 1990; Gordon et al. 2017). Walking is defined as a gait that lacks an aerial phase, as one foot is always in contact with the ground, and often occurs at low locomotor speeds. It is traditionally described by an RM known as the *inverted pendulum model* (IPM) (Saunders et al. 1953; Alexander 1976; Mochon and Mcmahon 1980; Alexander 1981, 1989; Srinivasan and Ruina 2005; Geyer et al. 2006; Gordon et al. 2017). In contrast, running occurs typically at higher speeds and involves an aerial phase, in which no feet are in contact with the ground, and dynamics are traditionally described with an RM known as the two-dimensional SMM (Alexander 1981, 1989; Blickhan 1989; McMahon and Cheng 1990; He et al. 1991; Farley et al. 1993; Gordon et al. 2017).

During IPM walking, the CoM rotates around the foot over a non-compliant leg and, as such, remains at a constant distance from that contact point (Saunders et al. 1953; Alexander 1976; Mochon and Mcmahon 1980). In the IPM, the leg is considered massless ($m = 0$) and infinitely stiff ($k = +\infty$). During stance, the motion is symmetric with respect to mid-stance; in the first half, the CoM rises from its touch-down height to its mid-stance height, converting its kinetic energy into potential energy, and in the second half, the potential energy is converted back into kinetic energy and

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

the CoM lowers to its take-off height, usually considered equal to its touch-down height (Cavagna et al. 1963, 1977; Cavagna and Margaria 1966; Mochon and Mcmahon 1980). The stance phase of running shares the symmetry with respect to mid-stance seen in walking, but, in contrast with the CoM trajectory of walking, which reaches its highest point at mid-stance, in running the CoM reaches its lowest point at mid-stance, thanks to compliance of the limbs (Alexander 2003b; Gordon et al. 2017; Patek and Biewener 2018). During running, in the first half of the stance, the CoM lowers from its touch-down height to its mid-stance height and converts its kinetic and gravitational potential energy into elastic energy. In the second half of the stance, elastic energy is converted back into kinetic and gravitational energy and the CoM raises to its take-off height, usually identical to its touch-down height (Alexander 2003a; Geyer et al. 2005). The two-dimensional SMM allows for both aerial and stance phases, with a massless but compliant ($k < \infty$) leg rotating around the point of contact of the foot with the ground. After take-off, the CoM motion of the aerial swing phase is ballistic (Gordon et al. 2017). In contrast with the IPM, the SMM can accurately describe the trajectory of the CoM and the patterns of ground reaction forces elicited not only during running, but also during walking (Geyer 2005; Geyer et al. 2006).

Although RMs are themselves simple and describe the essential components of the locomotion of an animal, their mathematical solutions are not always simple (Alexander 2003a; Geyer et al. 2005). In fact, the solutions of the equations of motion of such systems might not admit solutions which can be written explicitly in terms of elementary functions (e.g., polynomials, trigonometric functions, exponentials, and logarithms). For example, although during flight phase the dynamics is ballistic and so equations can be explicitly solved, the dynamics of the SMM during stance phase cannot be integrated (Whittacker 1904) and so, we have to use numerical simulations or find approximations in order to explore the properties of those equations and of the dynamics that they describe. Initial attempts at finding approximations neglected gravity ($g = 0$) in order to keep the solutions tractable enough for interpretation (e.g., Schmitt and Holmes 2000; Ghigliazza et al. 2003). However, due to the lack of realism of this scenario, attempts were made to reincorporate gravity (Schwind and Koditschek 2000; Geyer 2001), with Geyer et al. (2005) achieving success with a (relatively) simple approximation of the full system dynamics. Among the many insights that they can provide, approximate models and simulation studies have been useful for the study of gait stability. For example, people have shown that the SMM admits asymptotically stable gait trajectories

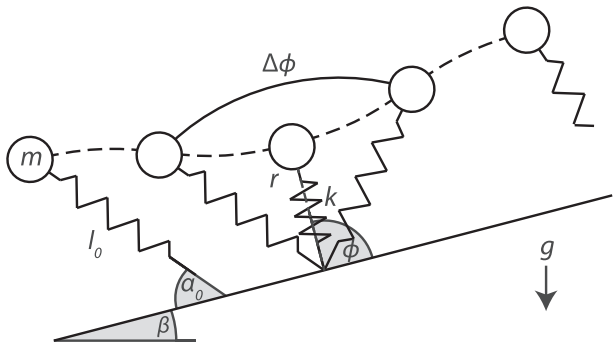


**Fig. 1** In this figure, $l_0$ represents the length of the leg at rest, $k$ is the stiffness of the spring, $m$ the mass of the animal, $\alpha_0$ is the angle of attack of the leg, $g$ is the constant of gravity, $\phi$ is the angle of the leg with respect to the inclined surface, $\Delta\phi$ is the angle swept during stance, $r$ is the radial distance of the mass from the contact point, and $\beta$ is the inclination of the surface.

(Ruina 1998; Coleman and Holmes 1999; Coleman et al. 2002; Seyfarth et al. 2002; Ghigliazza et al. 2003).

All of the studies mentioned above address the problem of modeling legged locomotion on level, uniform ground; much less is known about what mathematical equations describe locomotor behaviors of animals in more complex environments. The natural world is primarily composed of non-uniform surfaces, with variation in substrate incline, compliance, shape, and texture widely ubiquitous across terrestrial environments. Animals propelling themselves on these substrates thus have to compensate for these challenges, often trying to maintain some degree of locomotor performance in order to ensure survival, or at least successfully perform daily activities (Huey and Hertz 1982; Irschick and Jayne 1999; Schmidt and Fischer 2011; Birn-Jeffery and Higham 2014; Foster et al. 2015). Inclined surfaces, the focus of this manuscript, often induce changes in kinematics, such as decreased stride lengths and crouched postures, to increase stability (Jayne and Irschick 1999; Zaaf et al. 2001; Nakano 2002; Higham and Jayne 2004; Foster and Higham 2012; Birn-Jeffery and Higham 2014), and neuromuscular activity (Fowler et al. 1993; Carlson-Kuhta et al. 1998; Gillis and Biewener 2002; Gabaldón et al. 2004; Higham and Nelson 2008; Foster and Higham 2014, 2017), such as increased muscle fiber recruitment, to increase positive work. These insights have come from numerous studies that have examined these behavioral adjustments in live animals, but, unfortunately, modeling efforts have not kept pace with these experiments and, as a result, we lack a solid mathematical foundation for locomotion on inclines.

In this paper, we study the legged locomotion of animals undergoing stationary hopping, walking, and running on inclined surfaces, and mainly concentrate on the two-dimensional SMM (see Fig. 1) and its

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

relationship with other models, including the one-dimensional SMM and the IPM. Our main questions are: How do RMs change when hopping, walking, and running behaviors occur on inclines? Can we derive similar models and approximations for inclined locomotion as exist for level locomotion? What are the properties of these models and how well do they describe hopping, walking, and running gaits on inclines? To answer these questions, we analyze and compare one-dimensional spring-mass (stationary hopping), inverted pendulum (walking), and two-dimensional spring-mass (running) models on inclines, deduce approximate solutions, in particular for no gravity ($g = 0$) and small inclination ($0 < \beta$, but small), analyze the effect of the approximations on the solutions, discuss stability questions, perform simulations, and draw conclusions. We derive a simple approximate solution for the dynamics of the SMM on surfaces with small inclination ($0 < \beta$, but small) but including gravitational effects ($g > 0$). We provide some theoretical guarantees that are valid for small $\beta$ and provide insights valid for a full range of inclinations ($0 \leq \beta \leq \pi/2$). Our simulations consider model parameters relevant to human and bipedal bird (quail, pheasant, guinea fowl, turkey, and ostrich) locomotion and we also address the sensitivity with respect to surface inclination of important parameters such as mid-stance height, limb angle swept during stance, and stance phase contact time.

This paper is organized as follows. In the section "Lagrange formulation of reductionist models on inclines," we derive the Lagrangian formulations of the equations of motion of (i) one-dimensional SMM stationary hopping, (ii) IPM walking, and (iii) two-dimensional SMM running on inclines. In particular, we derive our approximate solution to the SMM and derive the formulas for the contact time, the angle swept during stance, and the mid-stance height predicted by our approximation. In the section "Approximate models and solutions with no gravity," we derive approximate models and solutions in the absence of gravity for (i) one-dimensional SMM stationary hopping, (ii) IPM walking, and (iii) two-dimensional SMM running on inclines. In the section "The effect of the inclination on the equations of motion," we discuss the effect of the inclination on the contact time, the angle swept during stance, and the mid-stance height. In the section "Comparisons of approximations," we discuss (i) the regimes under which the two-dimensional SMM reduces to the one-dimensional SM, (ii) the regimes under which the two-dimensional SMM reduces to the IPM, and (iii) the quality of our approximation of the solutions of the SMM. Section "Simulations" is dedicated to some simulations and the numerical analysis of the effect of the inclination in our model on the contact time, the angle swept during stance, and the mid-stance height in the running gait of six species (quail, pheasant, guinea fowl, turkey, ostrich, and human). Section "Discussion" is devoted to the discussion of our results, including questions of stability issues and related open problems, the limitations of our approximate solutions, and the possibility of transitions between behaviors. Section "Conclusions" is dedicated to our conclusions. In the Supplementary materials Appendix 1 "Mathematical Deductions" is dedicated to the mathematical deductions of our formulas, Appendix 2 "Theoretical Bounds" reports some theoretical bounds on our approximations, and Appendix 3 presents additional results related to the quality of our approximation across all six species.

## Lagrange formulation of reductionist models on inclines

In the following, $g$ is the gravitational constant and $m$ represents the mass of the body of the animal, while the massless spring has elastic constant $k$ and $\beta$ is the inclination of the surface. During flight phase (if any), all the models that we are considering behave ballistically, namely the acceleration of the mass is due only to the force of gravity. Given $y(t)$, the height from the surface of contact at time $t$, $\dot{y}(t)$, the vertical velocity at time $t$, and $\ddot{y}(t)$, the acceleration of the mass $m$ in the vertical direction at time $t$, we have $m\ddot{y}(t) + F_{gravity} = 0$, with $F_{gravity} = mg$ the force of gravity or weight of the mass. Given the initial conditions $y(0) = y_0$ and $\dot{y}(0) = v_0$, then the solution is

$$y(t) = -\frac{g}{2}t^2 + v_0 t + y_0.$$

If we assume that at the beginning and at the end of the stance phase the heights of the CoM are equal, then we have that the time of the aerial phase is given by $t_{aerial} = 2v_0/g$. During stance phase, the different models have different features and describe different dynamics, as we will discuss in the next subsections.

### Hopping—one-dimensional SMM

The one-dimensional SMM is relevant for understanding the dynamics of hopping on the spot, and it can be solved by a superposition of elementary functions during both contact and flight phases. The analysis of this subsection emerges from Gordon et al. (2017) and we report it here for completeness and comparison. During contact, the mass $m$ is influenced by both the gravitational force and the elastic force of the spring and so the dynamics of the CoM is given by

$$m\ddot{y}(t) + F_{spring} + F_{gravity} = 0,$$

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

which reduces to

$$m\ddot{y} + k(y - l_u) + mg = 0,$$

with $l_u$ being the length of the unloaded leg. Note that in this model, since the motion is purely vertical, the inclination of the surface $0 \leq \beta < \frac{\pi}{2}$ does not play a role. Defining the spring frequency $\omega_k = \sqrt{\frac{k}{m}}$ and including the initial conditions $y(0) = y_0$ and $\dot{y}(0) = v_0$, we obtain the general solution given by

$$y(t) = \frac{v_0}{\omega_k}\sin(\omega_k t) + \left(\frac{g}{\omega_k^2} + y_0 - l_u\right)\cos(\omega_k t) + l_u - \frac{g}{\omega_k^2}.$$

The mass $m$ oscillates around the height $h_g = l_0 - g/\omega_k^2$ with amplitude $A_g = \sqrt{(v_0/\omega_k)^2 + (g/\omega_k^2 - (l_u - y_0))^2}$. In the special case where the initial length of the spring equals its unloaded length, we get

$$y(t) = \frac{v_0}{\omega_k}\sin(\omega_k t) + \frac{g}{\omega_k^2}\cos(\omega_k t) + l_0 - \frac{g}{\omega_k^2}.$$

The motion is oscillatory around $l_0 - \frac{g}{\omega_k^2}$ with amplitude $A_g = \sqrt{(v_0/\omega_k)^2 + (g/\omega_k^2)^2}$.

### Walking—IPM

The IPM describes the walking gait through a differential equation of the angle $\phi(t)$ through which the leg rotates. The equation of motion of the CoM is given by

$$\ddot{\phi} = \frac{g}{l_0}\sin(\phi + \beta),$$

with $\ddot{\phi}$ representing the angular acceleration, $l_0$ being the length of the effective leg, and $\beta$ being the inclination of the surface. There are no explicit solutions to this model, but there are several ways to find approximate solutions. Assuming that we are interested in small oscillations with respect to the inclination, $|\phi - \pi/2| \ll 1$, we can approximate the IP as a simple harmonic oscillator:

$$\ddot{\phi} = \frac{g}{l_0}\cos(\beta),$$

whose solutions with initial conditions $\phi(0) = \phi_0$ and $\dot{\phi}(0) = \dot{\phi}_0$ are given by

$$\phi(t) = \frac{1}{2}\frac{g\cos(\beta)}{l_0}t^2 + \dot{\phi}_0 t + \phi_0.$$

### Running—two-dimensional SMM

The two-dimensional SMM is relevant for understanding the dynamics of running. It cannot be solved explicitly and so, to gain insight into the dynamics of the CoM under this model, approximations are needed. For consistency, we will use a notation similar to that in Geyer et al. (2005). The SMM on an inclined plane has the Lagrangian

$$L = L(r, \dot{r}, \phi, \dot{\phi}) = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{k}{2}(l_0 - r)^2 - mgr\sin(\phi + \beta)$$

and Lagrange equations given by

$$\begin{cases} m\frac{d}{dt}\dot{r} = mr\dot{\phi}^2 + k(l_0 - r) - mg\sin(\phi + \beta), \\ \frac{d}{dt}\left(mr^2\dot{\phi}\right) = -mgr\cos(\phi + \beta), \end{cases}$$

with boundary conditions at the beginning and end of the stance phase given by

$$\begin{cases} r_{TD} = r_{TO} \\ \dot{r}_{TD} = -\dot{r}_{TO} \\ \phi_{TO} = \phi_{TD} + \Delta\phi, \\ \dot{\phi}_{TO} = \dot{\phi}_{TD} \end{cases}$$

where $TD$ indicates touch-down conditions and $TO$ indicates take-off conditions. See Fig. 1 for more details about the parameters and the set-up of the system. In the flight phase, the trajectory reduces to parabolic. For sufficiently small angles during the stance phase $|\phi - \pi/2| \ll 1$, we can use the approximate Lagrangian $L_0$ given by

$$L_0 := L(r, \dot{r}, 0, \dot{\phi}) = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{k}{2}(l_0 - r)^2 - mgr\cos(\beta),$$

whose Lagrange equations are given by

$$\begin{cases} m\frac{d}{dt}\dot{r} = mr\dot{\phi}^2 + k(l_0 - r) - mg\cos(\beta), \\ \frac{d}{dt}\left(mr^2\dot{\phi}\right) = 0 \end{cases}$$

with the same boundary conditions for this model as for the original SMM. Consider the energy $E_0$ of the reduced model

$$E_0 := \frac{m}{2}\dot{r}^2 + \frac{P_0^2}{2mr^2} + \frac{k}{2}(l_0 - r)^2 + mgr\cos(\beta).$$

This energy can be rescaled (see Geyer et al. 2005) using the following changes of variables:

$$\tilde{\varepsilon} = \frac{2E_0}{ml_0^2}, \quad \omega = \frac{P}{ml_0^2}, \quad \omega_0 = \sqrt{\frac{k}{m}}, \quad \rho = \frac{r - l_0}{l_0}.$$

In this way, the *rescaled approximate energy* $\tilde{\varepsilon}$ becomes

$$\tilde{\varepsilon} = \dot{\rho}^2 + \frac{\omega^2}{(1+\rho)^2} + \omega_0^2\rho^2 + \frac{2g\cos(\beta)}{l_0}(1 + \rho).$$

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

A further approximation with small relative spring amplitude $|\rho| \ll 1$ (see Appendix 2) gives

$$\varepsilon = \dot{\rho}^2 + \omega^2(1 - 2\rho + 3\rho^2) + \omega_0^2\rho^2 + \frac{2g\cos(\beta)}{l_0}(1+\rho).$$

This equation can be solved explicitly by separation of variables. Transforming back the solution to the $r$ variable, we get

$$r(t) = l_0(1 + a + b\sin(\hat{\omega}t)),$$

with

$$a = \frac{\omega^2 - g\cos(\beta)/l_0}{\omega_0^2 + 3\omega^2},$$

$$b = \frac{\sqrt{(\omega^2 - g\cos(\beta)/l_0)^2 + (\omega_0^2 + 3\omega^2)(\epsilon - \omega^2 - 2g\cos(\beta)/l_0)}}{\omega_0^2 + 3\omega^2},$$

$$\hat{\omega}_0 = \sqrt{\omega_0^2 + 3\omega^2}.$$

The solution $r(t)$ describes a harmonic oscillation centered at $l_0(1 + a)$ with amplitude $l_0 b$ and is valid only for the contact phase $0 \le r(t) \le l_0$. From the explicit expression of the approximate solution, we can derive many quantities of interest. Using the condition $r_0 = l_0$, we can identify the *times of touch-down* $t_{TD}$ *and take-off* $t_{TO}$ by

$$t_{TD} = \frac{1}{\hat{\omega}_0}\left\{\left(2n + \frac{3}{2}\right)\pi - \left[\frac{\pi}{2} + \arcsin -\frac{a}{b}\right]\right\}$$

and

$$t_{TO} = \frac{1}{\hat{\omega}_0}\left\{\left(2n + \frac{3}{2}\right)\pi + \left[\frac{\pi}{2} + \arcsin -\frac{a}{b}\right]\right\},$$

with $n \in \mathbb{N}$. The *maximum spring compression* $\Delta l_{max}$ is given by the difference of the amplitude $l_0 b$ of $r(t)$ and the shift $l_0 a$ of the touch-down position $r(t_{TD})$. In terms of $a$ and $b$, the condition $|\rho| \ll 1$ is equivalent to $b - a \ll 1$. The *angle swept during stance* can be derived from the conservation of the angular momentum equation $P = mr^2\dot{\phi}$. In terms of the variables $\omega$ and $\rho$, the angular velocity is given by

$$\dot{\phi} = \frac{\omega}{(1+\rho)^2}.$$

Using the Taylor expansion to the first order, we get

$$\dot{\phi} = \omega(1 - 2\rho),$$

and substituting $\rho$ with $r$, we obtain the angle $\Delta\phi$ swept during stance

$$\Delta\phi = \int_{t_{TD}}^{t_{TO}} \omega\,[(1-2a) - 2b\sin(\hat{\omega}t)]\,dt = 2\frac{\omega}{\hat{\omega}_0}\left[(1-2a)\arccos\frac{a}{b} + 2\sqrt{b^2 - a^2}\right].$$

In our approximation, both the energy and the angular momentum are conserved and so $\Delta\phi$ is uniquely determined by the system state at touch-down $(l_0, \dot{r}_{TD}, \phi_{TD}, \dot{\phi}_{TD})$, the parameters of the spring-mass system $(k, m, g)$, and the inclination of the surface $\beta$. By fixing $t_{TD} = 0$, the radial and angular motions of the approximate solution are given by

$$r(t) = l_0 + l_0[a(1 - \cos(\hat{\omega}_0 t)) - \sqrt{b^2 - a^2}\sin(\hat{\omega}_0 t)]$$

and

$$\phi(t) = \phi_{TD} + (1-2a)\omega t + 2\frac{\omega}{\hat{\omega}_0}\left[a\sin(\hat{\omega}_0 t) + \sqrt{b^2 - a^2}(1 - \cos(\hat{\omega}_0 t))\right],$$

for $t \in [0, t_c]$ with $t_c = t_{TO} - t_{TD} = t_{TO} = \frac{1}{\hat{\omega}_0}\left\{\pi + 2\arcsin -\frac{a}{b}\right\}$ and $\phi_{TD} = \pi - \alpha_0$. Rewriting everything, including the expressions of $a$ and $b$, with respect to the parameters at touch down, we get

$$r(t) = l_0 - \frac{|\dot{r}_{TD}|}{\hat{\omega}_0}\sin(\hat{\omega}t) + \frac{\dot{\phi}_{TD}^2 - g\cos(\beta)}{\hat{\omega}_0^2}(1 - \cos(\hat{\omega}_0 t))$$

and

$$\phi(t) = \pi - \alpha_0 + \left(1 - 2\frac{\dot{\phi}_{TD}^2 - g\cos(\beta)/l_0}{\hat{\omega}_0^2}\right)\dot{\phi}_{TD}t + 2\frac{\dot{\phi}_{TD}}{\hat{\omega}_0}\left[\frac{\dot{\phi}_{TD}^2 - g\cos(\beta)/l_0}{\hat{\omega}_0^2}\sin(\hat{\omega}t) + \frac{|\dot{r}_{TD}|}{\hat{\omega}_0 l_0}(1 - \cos(\hat{\omega}_0 t))\right].$$

The radial motion corresponds to the motion of a one-dimensional SMM under the influence of gravity except for the increased oscillation frequency $\hat{\omega}_0 = \sqrt{k/m + 3\dot{\phi}_{TD}^2}$. The angular motion has a linear characteristic, which is modulated by trigonometric functions. The time spent during the stance phase is then given by

$$t_c = \frac{1}{\hat{\omega}_0}\left(\pi + 2\arctan\left\{\frac{g\cos(\beta) - l_0\dot{\phi}_{TD}^2}{|\dot{r}_{TD}|\hat{\omega}_0}\right\}\right).$$

## Approximate models and solutions with no gravity

As mentioned in the Introduction, approximations with $g = 0$ have been used extensively in the biomechanical literature. We discuss them here in the

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

context of inclines and see that when the no-gravity assumption is added to our approximation, the effect of the incline is not seen. Note that below there is a dual relationship between $\beta = \pi/2$ and $g = 0$, two cases that produce identical approximations under our assumptions.

### Hopping—one-dimensional SMM

In the case of no gravity ($g = 0$), the one-dimensional SMM simplifies and the dynamics of the CoM is given by

$$m\ddot{y}(t) + F_{spring} = 0,$$

which reduces to

$$m\ddot{y} + k(y - l_u) = 0,$$

with a general solution given by

$$y(t) = \frac{v_0}{\omega_k}\sin(\omega_k t) + \left(y_0 - l_u\right)\cos(\omega_k t) + l_u,$$

and in the special case $y_0 = l_0 = l_u$, we get

$$y(t) = \frac{v_0}{\omega_k}\sin(\omega_k t) + l_0.$$

The motion of the CoM is oscillatory as in the case $g > 0$, but in the case $g = 0$, the oscillation takes place around a higher center, namely $l_0$, which is bigger than $l_0 - g/\omega_k^2$, the height of the center of oscillation in the case for $g > 0$. The amplitude of the oscillation in the case $g > 0$ is higher/lower than in the absence of gravity ($g = 0$), if the following is satisfied/not satisfied :

$$g/\omega_k^2 > 2(l_u - y_0),$$

which is always satisfied in the case $y_0 = l_u = l_0$.

### Walking—IPM

The IPM without gravity ($g = 0$) reduces to an angular motion, which is linear in time:

$$\ddot{\phi} = 0,$$

with solution given by

$$\phi(t) = \phi_0 + \dot{\phi}_0 t.$$

### Running—two-dimensional SMM

In the absence of gravity ($g = 0$), the two-dimensional SMM on an inclined plane has the Lagrangian

$$L = L(r, \dot{r}, \phi, \dot{\phi}) = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{k}{2}(l_0 - r)^2$$

and Lagrange equations given by

$$\begin{cases} m\frac{d}{dt}\dot{r} = mr\dot{\phi}^2 + k(l_0 - r), \\ \frac{d}{dt}\left(mr^2\dot{\phi}\right) = 0 \end{cases}$$

with the same boundary conditions, as for $g > 0$. This regime does not need the small angle approximation $|\phi - \pi/2| \ll 1$, as the Lagrange equations do not depend on the angle explicitly. In this regime, the energy becomes identical to that of the reduced model for small angles, but with $g = 0$. Without need of any further approximation in this case, we obtain

$$E := \frac{m}{2}\dot{r}^2 + \frac{P_0^2}{2mr^2} + \frac{k}{2}(l_0 - r)^2.$$

Under changes of variables similar to that performed in the case $g > 0$, this energy can be rescaled to

$$\tilde{\varepsilon} = \dot{\rho}^2 + \frac{\omega^2}{(1+\rho)^2} + \omega_0^2\rho^2,$$

and with the further approximation for small relative spring amplitude $|\rho| \ll 1$, we get

$$\varepsilon = \dot{\rho}^2 + \omega^2(1 - 2\rho + 3\rho^2) + \omega_0^2\rho^2.$$

The solution in the $r(t) = l_0 + l_0\rho(t)$ variable again becomes

$$r(t) = l_0(1 + a + b\sin(\hat{\omega}t)),$$

with

$$a = \frac{\omega^2}{\omega_0^2 + 3\omega^2},$$

$$b = \frac{\sqrt{(\omega^2)^2 + (\omega_0^2 + 3\omega^2)(\epsilon - \omega^2)}}{\omega_0^2 + 3\omega^2},$$

and

$$\hat{\omega}_0 = \sqrt{\omega_0^2 + 3\omega^2}.$$

The solution $r(t)$, which is valid only for the contact phase $0 \le r(t) \le l_0$, describes a harmonic oscillation centered at $l_0(1 + a)$, which is now higher than in the case $g > 0$ with amplitude $l_0 b$. The amplitude of the oscillation is higher/lower in the absence of gravity, if the following is satisfied/not satisfied: $g\cos(\beta)/l_0 > 8\omega^2 + 2\omega_0^2$. Using the condition $r_0 = l_0$, and fixing $t_{TD} = 0$, the time spent in contact with the incline in the case $g = 0$ resolves to

$$t_c = \frac{1}{\hat{\omega}_0}\left(\pi + 2\arctan\left\{-\frac{l_0\dot{\phi}_{TD}^2}{|\dot{r}_{TD}|\hat{\omega}_0}\right\}\right),$$

which, by monotonicity of the arctan function, is shorter than in the case with gravity $g > 0$. Similar considerations can be made to analyze the maximum spring compression and the angle swept during stance.

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

## The effect of the inclination on the equations of motion

*Contact time*. As computed above, the time spent during the stance phase is given by

$$t_c = \frac{1}{\hat{\omega}_0}\left(\pi + 2\arctan\left\{\frac{g\cos(\beta) - l_0\dot{\phi}_{TD}^2}{|\dot{r}_{TD}|\hat{\omega}_0}\right\}\right).$$

By this formula, we can see that the contact time $t_c(\beta > 0) > t_c(\beta = 0)$. In fact, $\cos(\beta)$ is decreasing in $\beta$ for $0 < \beta < \pi/2$, and, as the arctan function is monotone in its argument, $t_c$ decreases for $\beta$ increasing inside the interval $0 < \beta < \pi/2$. *Angle swept during stance*. Consider the angle swept during stance

$$\Delta\phi = 2\frac{\omega}{\hat{\omega}_0}\left[(1-2a)\arccos\frac{a}{b} + 2\sqrt{b^2-a^2}\right].$$

We want to understand how this quantity varies with respect to the inclination $\beta$ of the surface. After some cumbersome computations (see Appendix 1), we have

$$\begin{aligned}\partial_\beta\Delta\phi = &-2\frac{\omega g\sin(\beta)}{\hat{\omega}_0^3 l_0} * 2\arccos \\ &\times \frac{\omega^2 - g\cos(\beta)/l_0}{\sqrt{(\omega^2 - g\cos(\beta)/l_0)^2 + (\hat{\omega}_0^2)\dot{r}_0^2/l_0^2}} \\ &-2\frac{\omega g\sin(\beta)}{\hat{\omega}_0^3 l_0} * \frac{(\omega_0^2+\omega^2)+2g\cos(\beta)/l_0}{\sqrt{(\omega^2-g\cos(\beta)/l_0)^2+(\hat{\omega}_0^2)\dot{r}_0^2/l_0^2}}.\end{aligned}$$

Therefore, $\partial_\beta\Delta\phi < 0$ and so the angle swept during stance decreases with increasing surface inclination for every $\beta \in [0, \pi/2]$. *Relative mid-stance height*. Consider again the approximated energy

$$\varepsilon = \dot{\rho}^2 + \omega^2(1-2\rho+3\rho^2) + \omega_0^2\rho^2 + \frac{2g\cos(\beta)}{l_0}(1+\rho),$$

at mid-stance $\rho = \rho_{mid}$ and $\dot{\rho}_{mid} = 0$ and so, by solving a second order equation in $\rho_{mid}$ and taking the derivative with respect to $\beta$, we obtain

$$\begin{aligned}\partial_\beta\rho_{mid} &= \frac{g\sin(\beta)/l_0 + \frac{(\omega^2-g\cos(\beta)/l_0)g\sin(\beta)/l_0}{\sqrt{(\omega^2-g\cos(\beta)/l_0)^2+(\omega_0^2+3\omega^2)\dot{r}_0^2/l_0^2}}}{\omega_0^2+3\omega^2} \\ &= \frac{g\sin(\beta)/l_0}{\omega_0^2+3\omega^2}\{1 \\ &\quad + \frac{(\omega^2 - g\cos(\beta)/l_0)}{\sqrt{(\omega^2-g\cos(\beta)/l_0)^2+(\omega_0^2+3\omega^2)\dot{r}_0^2/l_0^2}}\Bigg\} > 0.\end{aligned}$$

## Comparisons of approximations

### Two- to one-dimensional SMM

As on horizontal surfaces, on an incline ($\beta > 0$), intuition suggests that the equations of motion of running must reduce to those of hopping, once the equations of motion of running are restricted by the condition of "no horizontal displacement" ($\dot{\phi}^2 = 0, \quad \phi(t) = \frac{\pi}{2} - \beta$). This intuition can be made rigorous (with formal mathematical deductions) by showing that the two-dimensional SMM reduces to the one-dimensional SMM under the condition of "no horizontal displacement" also on an incline. Consider again the two-dimensional SMM with Lagrangian

$$\begin{aligned}L = L(r,\dot{r},\phi,\dot{\phi}) = &\frac{m}{2}\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{k}{2}(l_0 - r)^2 \\ &- mgr\sin(\phi+\beta),\end{aligned}$$

and assume that

$$\dot{\phi}^2 = 0, \quad \phi(t) = \frac{\pi}{2} - \beta.$$

In this case, the Lagrangian simplifies to

$$L = L(r,\dot{r},\phi,\dot{\phi}) = \frac{m}{2}\dot{r}^2 - \frac{k}{2}(l_0 - r)^2 - mgr,$$

with $r$ in the vertical position, and so $r(t) = y(t)$. The equations of motion in this case are then those of the one-dimensional SMM describing hopping on the spot:

$$m\ddot{y} + k(y - l_u) + mg = 0.$$

This deduction is also valid for the special case of $\beta = 0$, namely in the case of a horizontal surface.

### Two-dimensional SMM to inverted pendulum

Also on an incline, the limit case of a two-dimensional SMM with infinite stiffness resembles the IPM. Consider the two-dimensional SMM with Lagrange equations given by

$$\begin{cases} m\frac{d}{dt}\dot{r} = mr\dot{\phi}^2 + k(l_0 - r) - mg\sin(\phi+\beta) \\ \frac{d}{dt}\left(mr^2\dot{\phi}\right) = -mgr\cos(\phi+\beta), \end{cases}$$

and assume that

$$\frac{k}{m} \to +\infty.$$

The first of the two equations can be rewritten as

$$\frac{d}{dt}\dot{r}/\frac{k}{m} = r\dot{\phi}^2/\frac{k}{m} + (l_0 - r) - mg\sin(\phi+\beta)/\frac{k}{m},$$

which in the limit case imposes

$$r(t) = l_u.$$

If we substitute this in the second equation, we obtain

$$\frac{d}{dt}\left(ml_u^2\dot{\phi}\right) = -mgl_u\cos(\phi+\beta),$$

which simplifies to

$$\ddot{\phi} = -\frac{g}{l_u}\cos(\phi+\beta).$$

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

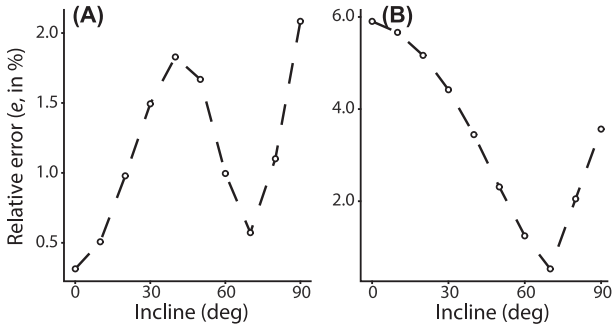


**Fig. 2** Relative error (e) of the radial distance of the CoM for our approximation ($r_{approx}$) **(A)** with gravity and **(B)** without gravity, expressed as a percentage. Parameters used were based on human values obtained from Iida et al. (2008); and Birn-Jeffery et al. (2014).

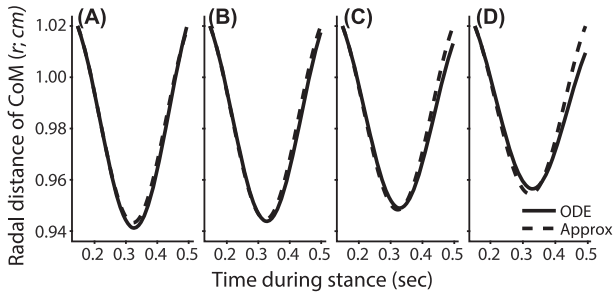


**Fig. 3** Radial distance of the CoM computed using the ODE ($r_{true}$; solid line) and our approximation ($r_{approx}$; dashed line) at **(A)** 0 deg, **(B)** 10 deg, **(C)** 20 deg, and **(D)** 30 deg. Parameters used were based on human values obtained from Iida et al. (2008); and Birn-Jeffery et al. (2014).

The equation of motion in this case is that of the IPM upon redefining the angle $\phi$ as the angle with respect to the surface instead of the angle with respect to the vertical. As shown in the reduction of the two-dimensional SMM to one-dimensional SMM above, this deduction is also valid for the special case of $\beta = 0$, namely in the case of a horizontal surface.

## Quality of our approximation to the two-dimensional SMM

In deducing our approximate solution for the SMM (for this paragraph, we will refer to it as $r_{approx}$, and to the numerical solution of the exact system with $r_{true}$), we made many simplifications, including those present in Geyer et al. (2005) $|\rho| \ll 1$, $|\phi| \ll 1$ and new ones $0 < \beta \ll 1$ related to the extra complexity required to model bipedal locomotion on inclines. A natural question is: How close is the approximate solution $r_{approx}$ to the true solution $r_{true}$ of the system? We will concentrate on the case of the SMM and human parameters. We measure this closeness in terms of the relative error $e$ of the $L^\infty$ norm (also called sup-norm), namely we calculate $e$ as

$$e := \frac{\|r_{approx}(t) - r_{true}(t)\|_{L^\infty[t_{TD},t_{TO}]}}{\|r_{true}(t)\|_{L^\infty}},$$

where $\|f(t)\|_{L^\infty[a,b]} := \sup_{t\in[a,b]} |f(t)|$. The $L^\infty$ norm measures the maximal error that the approximate solutions make in the interval of time considered. In Fig. 2, we show how $e$ varies with $\beta$ in the case in which we use parameters typical of human locomotion for gravity (A) and without gravity (B). We computed $r_{true}$ numerically, using the R-package *deSolve* and the available *ode* function. Plots of $r_{true}$ and $r_{approx}$ during stance at 0, 10, 20, and 30 deg (with gravity) are in Fig. 3. As you can see in Fig. 2A, the error made remains under 2% across all inclines, and it is extremely precise for small $\beta$ (e.g., $\beta = 0$ an error of $e = 0.3\%$ and $\beta = 10$ deg an error of $e = 0.5\%$). In contrast, when gravity is removed, along with the influence of incline (see the section "Running—two-dimensional SMM"), the error is greater at nearly every incline (Fig. 2B). Note that important locomotor parameters, such as the angle of attack $\alpha_0$, change in real situations while an animal moves on surfaces with different inclinations, whereas this computation of the relative error, $e$ assumes that these parameters are independent of $\beta$. The parameters we have used for the simulations are estimated parameters found in the literature and valid for the flat case $\beta = 0$. Therefore, although our error estimates are very good also for large $\beta$, our result for large $\beta$ needs to be taken with caution, considering also that for large $\beta$ our theoretical bounds are vacuous (see Appendix 2). Unfortunately, to the best of our knowledge, corresponding parameters for inclined surfaces across the range from 0 to 90 deg are not available in the literature.

## Simulations

To evaluate our model and the impact of inclination on key stance characteristics, we simulated stance phases for humans and five species of bipedal birds (quail, pheasant, guinea fowl, turkey, and ostrich) using parameters obtained from the literature. Specifically, values for $m$, $L_0$, $k$, and running speed ($v$) were obtained from Müller et al. (2016) and the references cited therein (Blum et al. 2011; Birn-Jeffery et al. 2014; Table 1). Running speed ($v$), also obtained from these sources, was used to determine reasonable stance contact times ($t_c$) based on published data for the quail (Gatesy and Biewener 1991), pheasant (Zhang et al. 2019), guinea fowl (Gatesy and Biewener 1991), turkey (Gatesy and Biewener 1991), ostrich (Rubenson et al. 2004), and human (Iida et al. 2008) (Table 1). Values for $t_c$ were used to identify appropriate values of $\varepsilon$ (or equivalently $\dot{r}_0/l_0$) and $\omega$ in our approximations. Using these parameters, we simulated the radial position of the center of mass ($r$) during stance phase by explicitly

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

**Table 1** Biological parameters for sensitivity analysis/simulations from Müller et al. (2016) and references therein.

| Speciies | $m$ (kg) | $L_0$ (m) | $k_{norm}$ | $v$ (m/s) | $t_c$ ($s^{-1}$) |
|---|---|---|---|---|---|
| Human[a] | 77 | 1.02 | 21.5 | 2 | 0.35 |
| Quail[b] | 0.2 | 0.117 | 8 | 1.2 | 0.09 |
| Pheasant[a] | 1.2 | 0.21 | 8.8 | 2.22 | 0.18 |
| Guinea Fowl[b] | 1.48 | 0.228 | 15 | 2.2 | 0.18 |
| Turkey[b] | 2.96 | 0.287 | 10 | 2.2 | 0.24 |
| Ostrich[b] | 116 | 0.974 | 12 | 3.2 | 0.33 |

[a]Blum et al. (2011)
[b]Birn-Jeffery et al. (2014)
In the table: $m$ is the mass, $L_0$ is the leg length, $k_{norm} := kl_0/(mg)$ is the normalized stiffness, $v$ is the running speed, and $t_c$ is the stance phase contact time approximated using $v$ and stance duration–velocity curves obtained for the human (Iida et al. 2008), quail (Gatesy and Biewener 1991), pheasant (Zhang et al. 2019), guinea fowl (Gatesy and Biewener 1991), turkey (Gatesy and Biewener 1991), and ostrich (Rubenson et al. 2004).

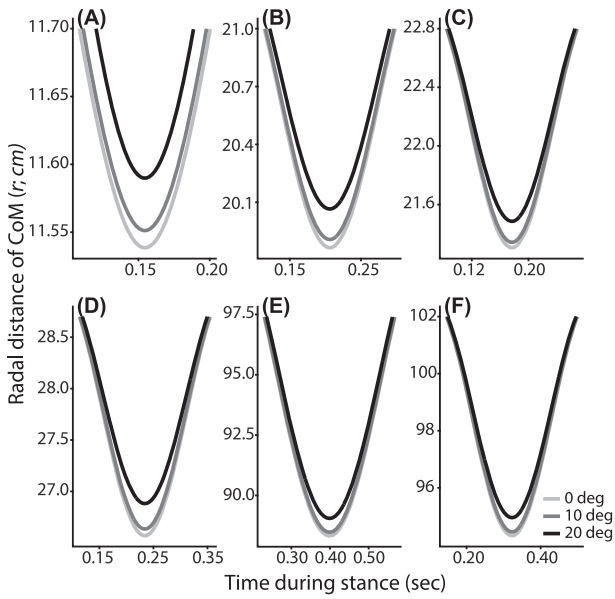


**Fig. 4** Radial distance of the CoM ($r_{approx}$) during stance phase using parameters relevant for **(A)** quail, **(B)** pheasant, **(C)** guinea fowl, **(D)** turkey, **(E)** ostrich, and **(F)** humans. Light gray, dark gray, and black represent 0, 10, and 20 deg inclines, respectively. Parameters for calculations obtained from Birn-Jeffery et al. (2014).

solving

$$r(t) = l_0(1 + a + b\sin(\hat{\omega}t)),$$

with $a$, $b$, and $\hat{\omega}_0$ defined as above. This was done for three inclines (0 , 10, and 20 deg) for all six species (see Fig. 4). In these simulations, $t_c$ and the range of values of $r$ during the course of the stance decrease (due to an increase in the minimum value of $r$ at mid-stance) as incline increases (Fig. 4). Further, the impact of incline on $r$ is greater in the larger species (Fig. 4).

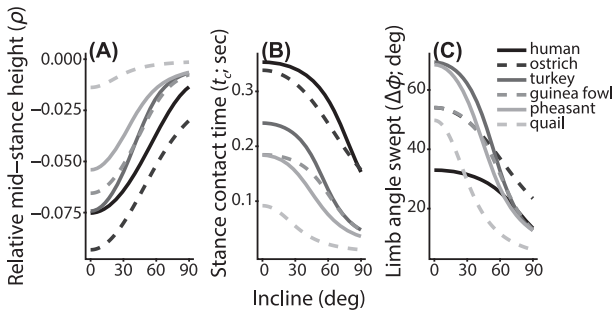


**Fig. 5** Dependence on incline of various stance characteristics using parameters relevant for quail, pheasant, guinea fowl, turkey, ostrich, and human. **(A)** limb compression at mid-stance (smaller $\rho$ is greater compression), **(B)** stance phase contact time (s), and **(C)** angle of the limb swept during stance ($\Delta\phi$). Parameters for calculations obtained from Birn-Jeffery et al. (2014).

To understand how contact time ($t_c$), angle swept during stance ($\Delta\phi$), and mid-stance height ($\rho_{mid}$) vary with incline ($\beta$) across these same six species, we again used the parameters listed above (Table 1) and the equations in the "The effect of the inclination on the equations of motion" section. The results of these simulations (Fig. 5) were consistent with those shown in Fig. 4. $\rho_{mid}$ increased with increasing incline and decreased as body size increased and the magnitude of the effect of incline on $\rho_{mid}$ increased with increasing body size (Fig. 5A). $t_c$ decreased with increasing incline and with decreasing body size, and, again, the magnitude of the impact of incline on $t_c$ increased with increasing body size (Fig. 5B). Finally, $\Delta\phi$ decreased with increasing incline, and, unlike for the other two variables, the impact of incline on $\Delta\phi$ was smaller in the two largest species (human and ostrich) than in the four smaller species (turkey, guinea fowl, pheasant, and quail; Fig. 5C).

## Discussion

### SMM on an inclined plane—simulations for bipedal runners

Our simulations using parameters relevant for bipedal humans and birds reveal several kinematic adjustments that are observed in animals dealing with inclined surfaces (Birn-Jeffery and Higham 2014). As incline increases, our simulations found that all species decrease their contact time and limb angle swept during stance (Figs. 4 and 5), although at 10–20 deg inclines, the effect on stance duration is negligible, which is consistent with literature for humans on shallow inclines (Gottschall and Kram 2005; Franz et al. 2012). The combination of these two effects is consistent with a decrease in stride length and increase in stride frequency, which have both been seen in kinematic studies of animals on inclines (Birn-Jeffery and Higham 2014). For example,

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

decreased stride length is observed in cats (Carlson-Kuhta et al. 1998), lizards (Jayne and Irschick 1999; Zaaf et al. 2001; Foster and Higham 2012), and turtles (Claussen et al. 2002), and increased stride frequency on inclines has been observed in humans (Minetti et al. 1994), other primates (Nakano 2002), and lizards (Irschick and Jayne 1998), though this latter change is by no means universal. Many species show little or no change in stride frequency and increase stance duration and duty factor on inclines (Irschick and Jayne 1998; Zaaf et al. 2001; Schmidt and Fischer 2011; Foster and Higham 2012). The scaling relationships that our model predicted for these variables are consistent with the literature as well; stride frequency tends to decrease with body size across all six species and stride length is larger in the four smaller species (quail, pheasant, turkey, and guinea fowl) than in the largest species (ostrich and human) (Gatesy and Biewener 1991; Daley and Birn-Jeffery 2018).

Our simulations also revealed a decrease in the range of radial distance of the CoM during stance phase (Fig. 4) and an increase in the relative height of the CoM at mid-stance (Fig. 5A) as incline increases. Together, this indicates that the leg undergoes less compression and the CoM oscillates less during stance as incline increases in models of our species. Reduced vertical oscillation of the hip is a common strategy employed by animals moving on steeper inclines (Vilensky et al. 1994; Carlson-Kuhta et al. 1998; Jayne and Irschick 1999), which is not surprising. Vertical oscillations of the CoM occur because of forces being applied perpendicular to the surface. As the surface inclination increases, a greater component of that perpendicular impulse acts to push the animal backwards away from the surface (Cartmill 1985; Birn-Jeffery and Higham 2014). A similar argument applies to the corresponding static situation, i.e., the magnitude of the CoM height. Animals running on steeper inclines tend to combat this increased toppling moment not only by decreasing oscillations of the CoM, but also by decreasing the overall magnitude of the CoM. They do this by adopting crouched postures through flexion of proximal and distal joints (Carlson-Kuhta et al. 1998; Spezzano and Jayne 2004; Schmidt and Fischer 2011; Foster and Higham 2012; Birn-Jeffery and Higham 2014). As our simple model does not have multiple joints, it is unable to simulate a crouched posture and thus we cannot replicate the crouched posture observed in animals. It is nevertheless interesting that the dynamic option, decreasing oscillations of the CoM, comes out as a natural consequence of our equations.

When comparing the results of our simulations in our six species, we generally found that the magnitude of the impact of incline on our variables increased as body size increased. Even on level surfaces, larger animals have to deal with the consequences of larger ground reaction forces, leading to morphological, behavioral, and biomechanical differences in animals of different body sizes (Biewener 1982, 2005). As the change in orientation of the surface relative to the direction at which gravity acts is the source of the toppling moments described above, and larger animals have larger masses upon which gravity acts, it is natural that the negative impacts of steeper inclines would increase with body size (Birn-Jeffery and Higham 2014). The single exception to this relationship in our simulations was the angle swept by the limb during stance, in which the decrease in this angle with incline was greater in the four smaller species (turkey to quail), but less when human and ostrich parameters were used. These two species are substantially (26–39 times) larger than the turkey and experience larger musculoskeletal demands (force, work, and power; Daley and Birn-Jeffery (2018)). To offset this cost, and reduce musculoskeletal strain, large birds run with smaller leg angles than smaller species, which decreases their relative stride length on horizontal surfaces (Daley and Birn-Jeffery 2018). On the steepest inclines, our model predicts that the difference between the leg angles of the different species decreases, with the larger species having slightly larger leg angles than the smaller species. This may be reasonable as ground reaction forces from the more massive bodies should be lower, reducing the amount of musculoskeletal strain experienced by these species on steep inclines. By starting off at smaller leg angles on horizontal surfaces, these larger species undergo less of an adjustment to reduce their leg angles to the values expected on the steepest inclines. It is interesting that our simple SMM predicts this size-dependent relationship in leg angle versus incline in the absence of directly incorporating factors related to strain of biological tissues or energetic costs.

## SMM on an inclined plane—stability analysis

There are many stability questions of importance for the SMM on inclines, some of which are a direct consequence of our analysis, while some others remain open. The stability of the exact solution with respect to initial data determines how the solution of a system of differential equations, such as the SMM, varies as a function of the initial conditions, including the initial radial position $r_0$, the initial radial velocity $\dot{r}_0$, the initial value of the angle $\phi_0$, and the initial angular velocity $\dot{\phi}_0$. Solutions of the SMM are stable in this sense, as the system of differential equations fits in the general mathematical theory addressing this problem, namely that of continuous dependence with respect to initial data of solutions of ordinary differential equations with regular

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

nonlinearities (Tao 2006). The stability of the approximate solutions with respect to the initial data is a simple consequence of the functional form of our approximate solutions. A similar discussion is valid for the stability of the true and approximate solutions with respect to changes in the parameters of our model because all of these appear as smooth functions in the equations of motion, the approximate equations, and the approximate solutions that we derived. The question of asymptotic stability of the gait trajectory, followed by a solution of the SMM, requires mathematical techniques that go beyond the scope of this paper, and so remains open. Note that the problem is non-trivial, as we are talking about stability of non-stationary periodic solutions. In the case of a flat surface ($\beta = 0$), this problem has been studied using the apex–return map (Geyer et al. 2005). The apex–return map is a discrete functional relationship $y_{i+1} = T_i(y_i)$, between the peaks of the motion across consecutive strides. The map $T$ depends on the equations of motion, the parameters of the model, and the initial conditions. Stationary solutions of this discrete map $y_{i+1} = y_i$, correspond to periodic solutions of the SMM on horizontal surfaces. These solutions are stable if the eigenvalues of $T$, as a map between appropriate functional spaces, are, in absolute value, less than 1: $\max |\lambda(T_i)| < 1$. Note that the question of asymptotic stability is also not completely understood in the horizontal case ($\beta = 0$), as Geyer et al. (2005) proved their asymptotic stability result in a relevant, but restricted, set of the parameter space, namely when the angular velocity at touch-down is identical to the pendulum frequency.

### Limitations and opportunities

Although our approximation is well-defined for every value of $\beta$ and provides insights in such cases, the original and approximate equations are asymptotically close for small values of $\beta$ and in such regimes, we have some theoretical guarantees (see Appendix 2). Nevertheless, our estimates of the errors between the true and approximate solutions for the radial distance of the CoM, reveal that the quality of our approximation is quite high across a wide range of inclines (e.g., 0.3, 0.5$, and 1% on 0, 10, and 20 deg inclines, respectively, for humans, and <4% for all species on inclines up to 30 deg; Figs. 2A and 3, and Supplementary Fig. S1A in Appendix 3). Even though the parameters used across all inclines were based on animals running on horizontal surfaces, and thus were not adjusted for kinematic alterations that would inevitably take place, the relative error reached a maximum at 2% on the vertical surface for humans (Fig. 2A). We would expect that the use of more relevant parameters, obtained from experimentation of animals moving across this entire range of inclines, would further reduce the error. Indeed, merely allowing angular velocity of the CoM ($\omega$) to change with incline further improves the approximation (Supplementary Fig. S1B).

An interesting property that comes out of our model is the ability to transition to new behaviors. At the mid-stance height, our model allows for the smooth transition to a horizontal dynamics of the CoM. Similar transitions have been observed during obstacle negotiation in birds where the fluctuation of the CoM diminishes and the animal adopts a more crouched posture on the obstacle (Daley et al. 2006). However, the SMM would constrain the motion to be stationary and so it cannot fully describe the dynamics after the transition. Furthermore, we conjecture that the height at which this transition is possible according to the SMM would be lower than the actual height at which this horizontal motion is biologically reasonable. This is because it has been shown that a crouched posture is energetically expensive and the spring component would naturally tend to push the animal vertically, at least to some extent. Further, the SMM is not complex enough to describe this behavior as this behavior would require bending the legs, which contradicts the assumptions of the SMM.

As animals run up inclines, they do work against the environment to achieve a net increase in the height of their CoM (Gabaldón et al. 2004; Birn-Jeffery and Higham 2014). Under the conditions under which we develop our approximation, the work against the environment is small in each stance phase ($W = 2{*}mgl_0{*}\cos(\alpha_0)\sin(\beta)$), and the stance phase dynamics can be relatively well-described (see Figs. 2 and 3 and Appendix 2 and 3) by our approximation. Conceptually, this situation is similar to approximating an almost constant curve with constant pieces at different heights over different intervals.

## Conclusions

In this paper, we have analyzed several aspects of the two-dimensional SMM on inclined surfaces and compared it with other reductionist models, including the one-dimensional SMM and the inverted pendulum on inclined surfaces.

Our analysis included the derivation of the Lagrange equations of motion of the (i) one-dimensional SMM for hopping, (ii) inverted pendulum for walking, and (iii) two-dimensional SMM on inclines. We derived an accurate and explicit approximate solution to the two-dimensional SMM on inclines and derived explicit formulas for the stance phase contact time, the angle swept by the limb during stance, and the mid-stance height predicted by our approximation. Furthermore, we derived approximate models and solutions in the

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

absence of gravity for the (i) one-dimensional SMM for hopping, (ii) inverted pendulum for walking, and (iii) two-dimensional SMM on inclines. We discussed (i) the regimes under which the two-dimensional SMM on inclines reduces to the one-dimensional SMM on inclines, (ii) the regimes under which the two-dimensional SMM on inclines reduces to the IPM on inclines, and (iii) the quality of our approximation of the solutions of the SMM on inclines with parameters typical of humans and bipedal birds. We performed simulations and numerical analyses to understand the effect of inclination on the contact time, the angle swept during stance, and the mid-stance height in our model for six species (quail, pheasant, guinea fowl, turkey, ostrich, and human). We discussed stability issues and the limitations of our approximate solutions and the possibility of transition between behaviors.

A comparison of the true and approximate solutions for the two-dimensional SMM on inclines using simulations based on parameters from bipedal birds and humans demonstrated the high-quality of the proposed approximation; relative error remained <4% for all species on inclines up to 30 deg (Figs. 2A and 3 and Supplementary Fig. S1A). These simulations revealed several important kinematic adjustments that are commonly observed in animals moving on inclines; simulations for all the species resulted in decreases in leg compression and angle swept during stance, indicating smaller vertical oscillations of the CoM and smaller stride lengths with increasing incline. Further, our simulations also indicate that the effects of incline increase as body size increases. The simplicity of extreme reductionist models, like the SMM, can be valuable tools that provide insight into the fundamental governing principles of locomotion. Their very simplicity facilitates interpretation and the thorough investigation of the principles they contain. Speculatively, expanding our understanding of the CoM dynamics of legged locomotion beyond uniform, horizontal surfaces is important not only for clarifying the mechanisms underpinning locomotion in the heterogeneous conditions of the natural world, but for elucidating the important properties and control mechanisms for proper functioning of artificial devices like robots. Our work highlights the contributions that mathematical modeling, and the corresponding computational analysis of such models, can provide to let important biological principles emerge.

## Author contributions

Both A.M.S. and K.L.F. were equally involved in the conception and design of the study, code development, running simulations, and writing and revising the manuscript. A.M.S. developed the mathematical equations. All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

## Acknowledgments

A.M.S. and K.L.F. dedicate this manuscript to the memory of Iolanda Buonfiglio and Suzanne Lois Foster, and thank their families for their constant support. They also thank the following funding sources for supporting the symposium: Society for Integrative and Comparative Biology, Micro Photonics Inc., and the Company of Biologists Scientific Meeting Grant.

## Funding

This work was supported by the NSF-Simons Pilot Project Program [grant number SP0046087 to A.M.S. and K.L.F]; Purdue University Fort Wayne start-up funds awarded to A.M.S.; and Ball State University start-up funds awarded to K.L.F.

## Supplementary data

Supplementary material is available at the *ICB* online.

## Data availability statement

Data used for simulations were obtained from publicly available sources as outlined in the paper.

## References

Alexander RM. 1976. Mechanics of bipedal locomotion. In: Davies PS, ed. Perspectives in experimental biology. Oxford: Pergamon Press. p. 493–504.

Alexander RM. 1981. The gaits of tetrapods: adaptations for stability and economy. Sym Zool Soc Lond 48:269–87.

Alexander RM. 1989. Optimization and gaits in the locomotion of vertebrates. Physiol Rev 69:1199–227.

Alexander RM. 2003a. Modelling approaches in biomechanics. Phil Trans R Soc Lond Ser B: Biol Sci 358:1429–35.

Alexander RM. 2003b. Principles of animal locomotion. Princeton (NJ): Princeton University Press.

Biewener AA. 1982. Bone strength in small mammals and bipedal birds: do safety factors change with body size? J Exp Biol 98:289–301.

Biewener AA. 2005. Biomechanical consequences of scaling. J Exp Biol 208:1665–76.

Birn-Jeffery AV, Higham TE. 2014. The scaling of uphill and downhill locomotion in legged animals. Int Comp Biol 54:1159–72.

Birn-Jeffery AV, Hubicki CM, Blum Y, Renjewski D, Hurst JW, Daley MA. 2014. Don't break a leg: running birds from quail to ostrich prioritise leg safety and economy on uneven terrain. J Exp Biol 217:3786–96.

Blickhan R. 1989. The spring-mass model for running and hopping. J Biomech 22:1217–27.

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

Blum Y, Birn-Jeffery AV, Daley MA, Seyfarth A. 2011. Does a crouched leg posture enhance running stability and robustness? J Theor Biol 281:97–106.

Carlson-Kuhta P, Trank TV, Smith JL. 1998. Forms of forward quadrupedal locomotion. II. A comparison of posture, hindlimb kinematics, and motor patterns for upslope and level walking. J Neurophysiol 79:1687–701.

Cartmill M. 1985. Climbing. In:Hildebrand M, Bramble DM, Liem KF, Wake DB, eds. Functional vertebrate morphology. Cambridge (MA): Harvard University Press. p. 73–88.

Cavagna GA, Heglund NC, Taylor CR. 1977. Mechanical work in terrestrial locomotion: two basic mechanisms for minimizing energy expenditure. Am J Physiol 233:R243–61.

Cavagna GA, Margaria R, Saibene FP. 1963. External work in walking. J Appl Phys 18:1–9.

Cavagna GA, Margaria R. 1966. Mechanics of walking. J Appl Phys 21:271–8.

Claussen DL, Lim R, Kurz M, Wren K. 2002. Effects of slope, substrate, and temperature on the locomotion of the ornate box turtle, *Terrapene ornata*. Copeia 2002:411–8.

Coleman MJ, Chatterjee A, Ruina A. 2002. Motions of a rimless spoked wheel: a simple three-dimensional system with impacts. Dyn Stab Syst 12:139–59.

Coleman MJ, Holmes P. 1999. Motions and stability of a piecewise holonomic system: the discrete Chaplygin sleigh. Regul Chaotic Dyn 4:1–23.

Daley MA, Birn-Jeffery AV. 2018. Scaling of avian bipedal locomotion reveals independent effects of body mass and leg posture on gait. J Exp Biol 221:jeb152538.

Daley MA, Usherwood JR, Felix G, Biewener AA. 2006. Running over rough terrain: guinea fowl maintain dynamic stability despite a large unexpected change in substrate height. J Exp Biol 209:171–87.

Dickinson MH, Farley CT, Full RJ, Koehl MA, Kram R, Lehman S. 2000; How animals move: an integrative view. Science 288:100–6.

Farley CT, Blickhan R, Saito J, Taylor CR. 1991. Hopping frequency in humans: a test of how springs set stride frequency in bouncing gaits. J Appl Phys 71:2127–32.

Farley CT, Glasheen J, McMahon TA. 1993. Running springs: speed and animal size. J Exp Biol 185:71–86.

Foster KL, Collins CE, Higham TE, Garland TJ. 2015. Determinants of lizard escape performance: decision, motivation, ability, and opportunity. In: Cooper WEJ, Blumstein DT, eds. Escaping from predators: an integrative view of escape decisions. Cambridge (MA): Cambridge University Press. p. 287–321.

Foster KL, Higham TE. 2012. How forelimb and hindlimb function changes with incline and perch diameter in the green anole, *Anolis carolinensis*. J Exp Biol 215:2288–300.

Foster KL, Higham TE. 2014. Context-dependent changes in motor control and kinematics during locomotion: modulation and decoupling. Proc R Soc Biol B: Biol Sci 281: 20133331.

Foster KL, Higham TE. 2017. Integrating gastrocnemius force-length properties, *in vivo* activation, and operating lengths reveals how *Anolis* deal with ecological challenges. J Exp Biol 220:796–806.

Fowler EG, Gregor RJ, Hodgson JA, Roy RR. 1993. Relationship between ankle muscle and joint kinetics during the stance phase of locomotion in the cat. J Biomech 26:465–83.

Franz JR, Lyddon NE, Kram R. 2012. Mechanical work performed by the individual legs during uphill and downhill walking. J Biomech 45:257–62.

Full RJ, Koditschek DE. 1999. Templates and anchors: neuromechanical hypotheses of legged locomotion on land. J Exp Biol 202:3325–32.

Gabaldón AM, Nelson FE, Roberts TJ. 2004. Mechanical function of two ankle extensors in wild turkeys: shifts from energy production to energy absorption during incline versus decline running. J Exp Biol 207:2277–88.

Garland TJ, Losos JB. 1994. Ecological morphology of locomotor performance in squamate reptiles. In: Wainwright PC, Reilly SM, eds. Ecological morphology: integrative organismal biology. Chicago (IL): University of Chicago Press. p. 240–302.

Gatesy SM, Biewener AA. 1991. Bipedal locomotion: effects of speed, size and limb posture in birds and humans. J Zool 224:127–47.

Geyer H, Seyfarth A, Blickhan R. 2005. Spring-mass running: simple approximate solution and application to gait stability. J Theor Biol 232:315–328.

Geyer H, Seyfarth A, Blickhan R. 2006. Compliant leg behaviour explains basic dynamics of walking and running. Proc R Soc B: Biol Sci 273:2861–7.

Geyer H. 2001. Movement criterion of fast locomotion: mathematical analysis and neuro-biomechanical interpretation with functional muscle reflexes Diploma Thesis. [Pittsburgh (PA)]: Carnegie Mellon University.

Geyer H. 2005. Simple models of legged locomotion based on compliant limb behavior [PhD Thesis]. [Jena]: Friedrich-Schiller-Universität.

Ghigliazza RM, Altendorfer R, Holmes P, Koditschek DE. 2003. A simply stabilized running model. SIAM J Appl Dyn Syst 2:187–218.

Gillis GB, Biewener AA. 2002. Effects of surface grade on proximal hindlimb muscle strain and activation during rat locomotion. J Appl Phys 93:1731–43.

Gordon MS, Blickhan R, Dabiri JO, Videler JJ. 2017. Animal locomotion: physical principles and adaptations. Boca Raton (FL): CRC Press.

Gottschall JS, Kram R. 2005. Ground reaction forces during downhill and uphill running. J Biomech 38:445–52.

He JP, Kram R, McMahon TA. 1991. Mechanics of running under simulated low gravity. J Appl Phys 71:863–70.

Higham TE, Jayne BC. 2004. Locomotion of lizards on inclines and perches: hindlimb kinematics of an arboreal specialist and a terrestrial generalist. J Exp Biol 207:233–48.

Higham TE, Nelson FE. 2008. The integration of lateral gastrocnemius muscle function and kinematics in running turkeys. Zoology 111:483–93.

Holmes P, Full RJ, Koditschek D, Guckenheimer J. 2006. The dynamics of legged locomotion: models, analyses, and challenges. SIAM Rev 48:207–304.

Huey RB, Hertz PE. 1982. Effects of body size and slope on sprint speed of a lizard (*Stellio (Agama) stellio*). J Exp Biol 97: 401–9.

Iida F, Rummel J, Seyfarth A. 2008. Bipedal walking and running with spring-like biarticular muscles. J Biomech 41:656–67.

Irschick DJ, Jayne BC. 1998. Effects of incline on speed, acceleration, body posture and hindlimb kinematics in two species of lizard *Callisaurus draconoides* and *Uma scoparia*. J Exp Biol 201:273–87.

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026

Irschick DJ, Jayne BC. 1999. A field study of the effects of incline on the escape locomotion of a bipedal lizard, *Callisaurus draconoides*. Physiol Biochem Zool 72:44–56.

Jayne BC, Irschick DJ. 1999. Effects of incline and speed on the three-dimensional hindlimb kinematics of a generalized iguanian lizard (*Dipsosaurus dorsalis*). J Exp Biol 202:143–59.

Margaria R. 1976. Biomechanics and energetics of muscular exercise. Oxford: Clarendon Press.

McGeer T. 1990a. Passive dynamic walking. Int J Robot Res 9: 62–82.

McGeer T. 1990b. Passive bipedal running. Proc R Soc B: Biol Sci 240:107–34.

McMahon TA, Cheng GC. 1990. The mechanism of running: how does stiffness couple with speed? J Biomech 23:65–78.

Minetti AE, Ardigo LP, Saibene F. 1994. Mechanical determinants of the minimum energy cost of gradient running in humans. J Exp Biol 195:211–25.

Mochon S, Mcmahon TA. 1980. Ballistic walking. J Biomech 13:49–57.

Müller R, Birn-Jeffery AV, Blum Y. 2016. Human and avian running on uneven ground: a model-based comparison. J Roy Soc Int 13:20160529.

Nakano Y. 2002. The effects of substratum inclination on locomotor patterns in primates. Z Morphol Anthropol 83:189–99.

Patek SN, Biewener AA. 2018. Animal locomotion. Oxford: Oxford University Press.

Raibert M, Blankespoor K, Nelson G, Playter R. 2008. BigDog, the rough-terrain quadruped robot. IFAC Proc Vol 41: 10822–5.

Rubenson J, Heliams DB, Fournier PA. 2004. Gait selection in the ostrich: mechanical and metabolic characteristics of walking and running with and without an aerial phase. Proc R Soc B: Biol Sci 271:1091–9.

Ruina A. 1998. Non-holonomic stability aspects of piecewise holonomic systems. Rep Math Phys 42:91–100.

Saunders JDM, Inman VT, Eberhart HD. 1953. The major determinants in normal and pathological gait. J Bone Joint Surg 35A:543–728.

Schmidt A, Fischer MS. 2011. The kinematic consequences of locomotion on sloped arboreal substrates in a generalized (*Rattus norvegicus*) and a specialized (*Sciurus vulgaris*) rodent. J Exp Biol 214:2544–59.

Schmitt J, Holmes P. 2000. Mechanical models for insect locomotion: dynamics and stability in the horizontal plane I. Theory. Biol Cybern 83:501–15.

Schwind WJ, Koditschek DE. 2000. Approximating the stance map of a 2-DOF monoped runner. J Non Sci 10:533–68.

Seok S, Wang A, Chuah MY, Otten D, Lang J, Kim S. 2013. Design principles for highly efficient quadrupeds and implementation on the MIT Cheetah robot. 2013 IEEE International Conference on Robotics and Automation, Karlsruhe, Germany.

Seyfarth A, Apel T, Geyer H, Blickhan R. 2001. Limits of elastic leg operation. In: Blickhan R, ed. Motion systems. Aachen, Germany: Shaker Verlag. p. 102–7.

Seyfarth A, Geyer H, Günther M, Blickhan R. 2002. A movement criterion for running. J Biomech 35:649–55.

Spezzano LC, Jayne BC. 2004. The effects of surface diameter and incline on the hindlimb kinematics of an arboreal lizard (*Anolis sagrei*). J Exp Biol 207:2115–31.

Srinivasan M, Ruina A. 2005. Computer optimization of a minimal biped model discovers walking and running. Nature 439:72–75.

Tao T. 2006. Nonlinear dispersive equations: local and global analysis. In: Conference Board for the Mathematical Sciences. Providence (RI): American Mathematical Society.

Vilensky JA, Moore AM, Libii JN. 1994. Squirrel monkey locomotion on an inclined treadmill: Implications for the evolution of gaits. J Hum Evol 26:375–86.

Whittacker ET. 1904. A treatise on the analytical dynamics of particles and rigid bodies. Cambridge (MA): Cambridge University Press.

Zaaf A, Van Damme R, Herrel A, Aerts P. 2001. Spatio-temporal gait characteristics of level and vertical locomotion in a ground-dwelling and a climbing gecko. J Exp Biol 204: 1233–46.

Zhang R, Han D, Ji Q, Li G, Li X, Li J. 2019. Effects of speed on pheasant gaits and hindlimb joint angles. Anim Biol 69: 293–306.

Downloaded from https://academic.oup.com/icb/article/62/5/1320/6604836 by guest on 28 February 2026